

Phase 15 — Billing & Invoicing: Engineering Report

Status: COMPLETE **Date:** 2026-07-11 **PRs:** #113, #114, #115, #116, #117, #118, #119, #120, #121, #122, #123, #124, #125
ADRs: ADR-0035 (new); ADR-0034 (referenced, carve-out implemented in 15.7)

Sub-phase summary

Sub-phase	PR	Scope	Key finding
15.1	#113	Facility pricing (optional, paise, rupee UX)	Whole-number entry must not force .00; fixed proactively before first run
15.2	#114	Generation-time policy field	Mirrors the 4-field Policies fetch-on-mount pattern exactly
15.3	#115	Invoice generation engine, Cloud Scheduler	New keyless sa-scheduler-invoker; live-verified real invoices before merge
—	#117	Correction to 15.3	Denormalized flat_number/resident_name at generation time (avoids display-time N+1 at any scale); found a real “first resident, even if unresolvable” bug via live data, fixed same PR
15.4	#116	Resident’s own invoice view	Cross-household isolation tested both directions
15.4b	#118	Tenant-admin latest-per-flat	Zero profile lookups — reused 15.3’s denormalization
15.4c	#119	Per-flat history + live current-month preview	Extracted shared charge-computation function; 18 pre-existing tests confirmed unmodified after refactor
15.5	#120	CSV/JSON export, private GCS, manual triggers	Closed the long-flagged “15.3b” manual-generation gap; found and fixed the same-PR flat_number tie-break bug via live data before merge
—	#121	Deploy-wiring fix	SPORTSLOT_CLOUD_RUN_SA_EMAIL never reached the live container — found via a real UI click, not a test

—	#123	IAM fix	sa-cloud-run had signing rights but no actual bucket read/write — found via a real AccessDenied, not a test
15.6	#122	Agent invoice tools (read-only)	Surfaced the phase's most significant finding — see below
—	#124	Agent reliability fix	Deterministic pre-Vertex keyword routing; live-reproduced Gemini tool-selection non-determinism, confirmed via direct testing before any fix was designed
15.7	#125	ADR-0034 invoice-exclusion carve-out	Dynamic subcollection enumeration, not a hardcoded list — proven via a fixture subcollection appearing in no hardcoded list anywhere in the codebase

Metrics

- Backend: 542 tests, 92.55% coverage (gate: 90%)
- New Firestore collection: invoices (/tenants/{id}/invoices/{id}, deterministic ID {household_id}_{YYYY-MM}, create-only/immutable)
- New GCP infrastructure: sa-scheduler-invoker service account, Cloud Scheduler job (invoice-generation-monthly), private GCS bucket (sport-slot-dev-invoices), self-impersonation IAM grant for keyless signed URLs
- New backend routes: POST /internal/invoicing/generate (scheduler- only), GET /invoices/mine, GET /invoices/tenant/latest, GET /invoices/tenant/{household_id}/history, GET /invoices/tenant/{household_id}/preview, POST /invoices/tenant/regenerate, POST /invoices/tenant/export, signed-URL download route
- New agent tools: get_my_invoices, get_my_current_month_charges, plus a new deterministic pre-Vertex routing layer in run_agent

Real incidents found and resolved same-day (not carried over)

Five distinct real bugs, none caught by a passing test suite alone — every one found by insisting on live verification before calling a sub-phase done:

1. **flat_number locking onto an unresolvable first resident** — found by generating a real invoice against real (partially deleted) production data.
2. **Missing deploy-time env var** (SPORTSLOT_CLOUD_RUN_SA_EMAIL) — found via a real button click producing a real 500, traced to scripts/deploy_cloud_run.sh never being updated in the PR that added the corresponding config.py field.
3. **Missing GCS bucket IAM grant** — sa-cloud-run had the self-impersonation *signing* right but no actual object read/write permission on the new bucket; found via a real AccessDenied.
4. **Gemini tool-selection non-determinism** — the phase's most significant finding. Reproduced directly (identical phrasing, fresh sessions, one success one failure) before any fix was designed;

ruled out both a wording gap and a stale deployment as causes before concluding it was genuine model non-determinism.

5. **A same-timestamp confirmed/cancelled tie-break** (Daily Booking Overview, adjacent work same day) — caught by reading the actual dict-overwrite logic during review, not by any test that happened to exercise the exact collision.

Deliberately deferred (not incomplete — decided)

- **Per-tenant invoice generation time** — the Cloud Scheduler trigger is a single global fixed time; the tenant-configurable policy field (15.2) is stored and displayed but not yet wired into scheduling.
- **Invoice correction/adjustment mechanism** — invoices are immutable with no update path; a genuinely wrong invoice currently has no in-app remedy.
- **Generalizing deterministic pre-Vertex routing** to the other four agent tools — explicitly scoped narrowly to the two invoice tools this session; a real, separate future design question.

ADR-0035

New ADR covering all ten major architectural decisions from this phase: pricing model, billing cycle, invoice immutability, no payment tracking, household-level billing resolution, GCS export's role as a derivative (not the system of record), the two manual recovery triggers, tenant-admin visibility's evolution (latest-only → history + live preview), agent access and its reliability fix, and the ADR-0034 carve-out's actual implementation. Accepted at phase close, same discipline as ADR-0034 — every decision in it independently verified against live production behavior before being written up as accepted, not merged-and-assumed.

Carried into the next phases

- Voice input/output for the AI Booking Assistant (design locked, ADR to be written during that phase's own design session, matching this project's established practice)
- Backup & Disaster Recovery (ADR + implementation), sequenced after Voice I/O — this phase's own existence is the clearest argument yet for why it's needed: real financial data now exists with no backup strategy defined